PAVAN SRINIVAS DOMA

AI Engineer
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AI Engineer with a Master's in Engineering Science and over a year as a data engineer at the University at Buffalo. Skilled in AWS, RAG and SQL, with academic projects in retrieval-augmented conversational AI and credit scoring & customer segmentation. Built a RAG system with 97.5% semantic relevance accuracy, containerized on AWS ECS.

SKILLS

Languages: SQL, Python
Cloud & DevOps: AWS, ECS, EC2, Lambda, Containerization, Docker
LLMs & GenAI: RAG Systems, Fine-tuning, LLM Evaluation, LangGraph, Prompt Engineering, Vector Databases, Hugging Face
Data & Analytics: Data Pipelines, Matplotlib, NumPy, Pandas
Machine Learning: Natural Language Processing, Keras, PyTorch, Scikit-Learn

ACADEMIC PROJECTS

Retrieval-Augmented Conversational AI Systems

- Designed a retrieval-augmented generation (RAG) system and validated its semantic relevance at 97.5% accuracy with ML engineers; containerized it on AWS ECS with monitoring.
- Implemented context engineering, conversation memory and structured output validation so the assistant gave consistent, context-aware answers.

Credit Scoring & Customer Segmentation

- Compared 5 model approaches in MLflow with senior engineers and checked the business logic against data quality trade-offs.
- Developed a credit risk scoring model and defined its feature engineering strategy with a cross-functional team.

Consumer Complaint Classification

- Built an intent classifier for banking customer queries covering 77 intent classes, reaching 89.5% accuracy.
- Added a confidence threshold that let 49.6% of predictions skip manual review while keeping 99.8% accuracy on the auto-approved cases.

PROFESSIONAL EXPERIENCE

Data Engineer - Lead — Buffalo, USA
University at Buffalo — Sep 2024 - Dec 2025

- Improved ETL pipeline performance through SQL query refactoring and schema redesign, supporting real-time operational decisions that raised asset utilization efficiency by 25%.
- Standardized and documented ETL pipelines for stock tracking and allocation, adding data quality gates and error handling that cut data ingestion errors by 30%; wrote runbooks that helped onboard new engineers faster.
- Built a production resource allocation pipeline with SQL window functions and time-series decomposition, working with stakeholders to validate the business logic from extraction to the insights shared with leadership.
- Built real-time KPI dashboards in advanced SQL (window functions, CTEs, aggregations) and optimized the queries behind them, cutting query runtime by 40%.

EDUCATION

University at Buffalo, SUNY — Buffalo, USA
Master's in Engineering Science — Jan 2026

S.R.M Institute of Science and Technology, SRM University — Chennai, India
Bachelor of Technology in Artificial Intelligence — May 2024

CERTIFICATIONS

- Claude with Amazon Bedrock (Generative AI & LLM Application Development) — Jul 2026
- AWS Cloud Practitioner (Cloud platform deployment & AI services) — Jan 2026